

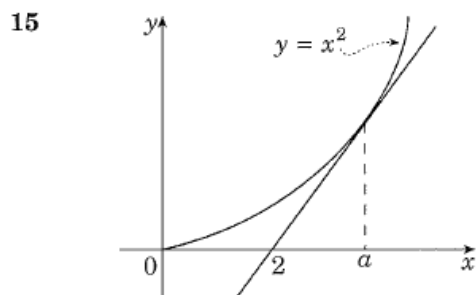
DIFFERENTIATION

Examination questions

- 1 Differentiate $5x^3 - 3$
- 2 Differentiate $3x^5 + x$
- 3 Differentiate $3x^4 - 2x + 3$
- 4 Differentiate $4x^3 - 2x + \frac{2}{x^2} + 3$
- 5 Differentiate $5x^2 - 8 + \frac{4}{x}$
- 6 Differentiate $\frac{1}{x^4}$
- 7 Differentiate $\frac{2}{x^3}$
- 8 Differentiate $3x^{-2}$
- 9 Differentiate $\sqrt{x}$
- 10 Differentiate $\frac{x}{2-x}$
- 11 Differentiate $(x-1)^{15}$
- 12 Differentiate $(x^2+1)^8$
- 13 Differentiate $3(2-x)^5$
- 14 Find the equation of the tangent to the curve

$$y = x^2 + \frac{2}{x} - 1$$

at the point $P(1, 2)$.



The diagram shows a graph of the parabola $y = x^2$ and the tangent to the parabola at $x = a$.

- (i) Find the gradient of the parabola at $x = a$.
 - (ii) Find equation of the tangent at $x = a$.
 - (iii) Find the value of a if the tangent intersects the x -axis at $x = 2$.
- 16 The line $y = mx + b$ is a tangent to the curve $y = x^3 - 3x + 2$ at the point $(-2, 0)$. Find m and b .
 - 17 Find the equation of the normal to the curve $y = x^2 - x$ at the point where $x = 2$.
 - 18 The curve $y = 3x^2 - \frac{p}{x^2}$ has a turning point at $x = -1$. Find the constant p .

- 19 The function $h(x)$ is defined by the rule

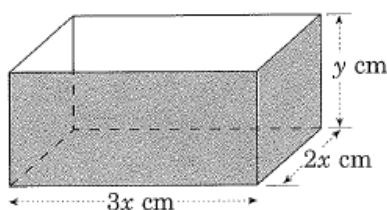
$$h(x) = (x-1)(x^2-5).$$

- (a) Find the real roots of the equation $h(x) = 0$.
 - (b) Find the coordinates of the turning points of $h(x)$, and state whether they are a maxima or minima.
 - (c) Draw a sketch of the graph $y = h(x)$ in the domain $-3 \leq x \leq 3$.
- 20 The function $g(x)$ is given by $g(x) = x(x+2)^2$.
 - (a) Find the coordinates of the points at which the graph of $y = g(x)$ meets the x -axis.
 - (b) Find the coordinates of the turning points of $g(x)$ and state whether they are maxima or minima.
 - (c) Draw a sketch of $y = g(x)$ in the domain $-3 \leq x \leq \frac{1}{2}$.
 - 21 Consider the function whose derivative is given by $\frac{dy}{dx} = x^2(x-2)(x+1)$. Determine the nature of the stationary point at $x = 0$.
 - 22 Consider the curve given by $y = 1 - 12x + x^3$ for $-4 \leq x \leq 4$.
 - (i) Find the stationary points and determine their nature.
 - (ii) Find the point of inflexion.
 - (iii) Sketch the curve for $-1 \leq x \leq 4$.
 - (iv) What is the minimum value of y for $-1 \leq x \leq 4$ and where does it occur?
 - 23 Consider the curve given by $y = 6x^2 - 2x^3$.
 - (i) Find the stationary points and determine their nature.
 - (ii) Sketch the curve, indicating where it intersects the x -axis.
 - (iii) Find the equation of the tangent to the curve at the point $P(1, 4)$.
 - 24 Consider the curve given by $y = \frac{1}{3}x^3 - x^2$.
 - (i) Find any turning points and determine their nature.
 - (ii) Find any points of inflexion.
 - (iii) Sketch the curve for the domain $-0.5 \leq x \leq 3.5$.
 - (iv) For what values of x is the curve concave down?

- 25 Sixteen koala bears were released from captivity into a wildlife park. The population P of koalas in the park t years later is given by

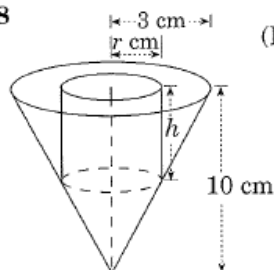
$$P = -t^3 + 6t^2 + 16, \text{ for } 0 \leq t \leq 5.$$

- After how many years was the population a maximum?
 - What was the maximum population?
 - Sketch the curve $P = -t^3 + 6t^2 + 16$ for $0 \leq t \leq 5$.
 - When was the population increasing most rapidly?
- 26 An open rectangular box has four sides and a base, but no lid. The figure below shows the box which has dimensions of $3x$ cm, $2x$ cm and a height of y cm.



- Write down formulae for the outer surface area A cm² of the box and the volume V cm³ contained by the box.
 - It is known that $A = 240$. Eliminate y to obtain a formula $V(x)$ for the volume as a function of x .
 - Show that $x < 2\sqrt{10}$.
 - Find the value of x for which V is a maximum and verify the maximum value of V is $64\sqrt{30}$.
- 27 An injured animal is given an antibiotic injection, which permeates into the animal's bloodstream over a period of time. The amount, A , of antibiotic present in the animal's bloodstream t hours later is given by
- $$A = 5t^2 - t^3, \text{ for } 0 \leq t \leq 4.$$
- Sketch the curve $A = 5t^2 - t^3$ for $0 \leq t \leq 4$, showing any stationary points.
 - At what time is the amount of antibiotic in the animal's bloodstream a maximum?
 - When is the amount present increasing most rapidly?

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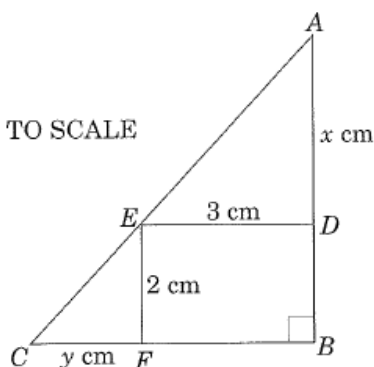
(Figure not to scale)

A cylinder of radius r cm and height h cm is inscribed in a cone with base radius 3 cm and height 10 cm, as in the diagram.

- Show that the volume V of the cylinder is given by $V = \frac{10\pi r^2(3-r)}{3}$.
- Hence find the values of r and h for the cylinder which has maximum volume.
- What is the maximum volume?

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NOT TO SCALE

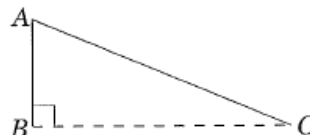


$DEFB$ is a rectangle with sides $ED = 3$ cm and $EF = 2$ cm. A and C lie on the lines BD and BF respectively, so that A , E and C are collinear as shown in the diagram.

Let $AD = x$ cm and $CF = y$ cm.

- Show that triangles ADE , EFC are similar.
 - Show that $xy = 6$.
 - Show that the area, A , of triangle ABC is given by $A = \frac{3x}{2} + \frac{6}{x} + 6$.
 - Find the height and base of triangle ABC with minimum area. Justify your answer.
- 30 A plane is to fly 3000 km at a constant speed of v km/h. When flying at v km/h the plane consumes fuel at the rate of $(50 + 10^{-6}v^3)$ litres per hour.
- Show that on a journey of 3000 km at a speed of v km per hour, the expression for the total amount of fuel used, A litres, is given by $A = \frac{150\,000}{v} + \frac{3v^2}{1000}$.
 - Find the speed for the greatest fuel economy and the amount of fuel used at this speed. Give both answers correct to three significant figures.

31



A piece of wire of length 5 metres is bent to form the hypotenuse and side of a right-angled triangle ABC , as shown in the diagram. Let the length of the side AB be x metres.

- What is the length of the hypotenuse AC in terms of x ?

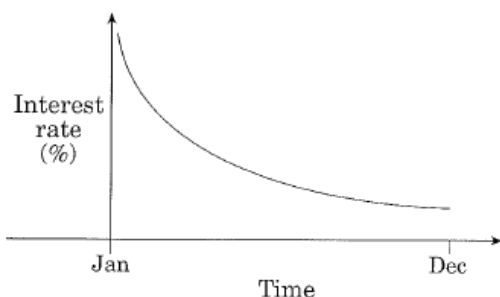
(ii) Show that the area of the triangle ABC is $\frac{1}{2}x\sqrt{25-10x}$ square metres.

(iii) What is the maximum possible area of the triangle?

- 32** 'Although the world's population continues to increase, the rate of population growth is decreasing.'

Draw a graph of population as a function of time that fits this description.

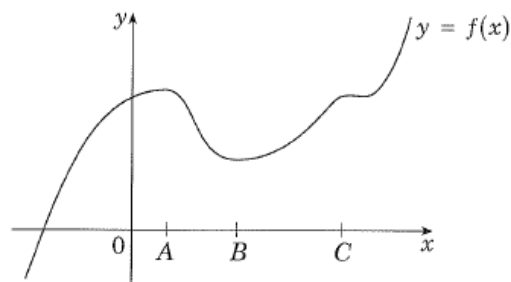
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The graph shows the interest rate for housing loans over a period of a year.

Describe briefly how the interest rate has changed over this period. Include mention of the rate of change of the interest rate.

- 34** The diagram shows the graph of a certain function $y = f(x)$. The graph has maximum and minimum turning points and a point of horizontal inflexion in the domain shown.



- (i) Copy this graph into your Writing Booklet.
(ii) On the same set of axes, draw a sketch of the derivative $f'(x)$ of the function.

- 35** For all x in the domain $0 \leq x \leq 3$, a function $f(x)$ satisfies $f'(x) < 0$ and $f''(x) < 0$.

Sketch a possible graph of $y = f(x)$ in this domain.

DIFFERENTIATION

Worked solutions to examination questions

1 $\frac{d}{dx}(5x^3 - 3) = 15x^2$

2 $\frac{d}{dx}(3x^5 + x) = 15x^4 + 1$

3 $\frac{d}{dx}(3x^4 - 2x + 3) = 12x^3 - 2$

4 $\frac{d}{dx}\left(4x^3 - 2x + \frac{2}{x^2} + 3\right) = \frac{d}{dx}(4x^3 - 2x + 2x^{-2} + 3)$
 $= 12x^2 - 2 - 4x^{-3}$
 $= 12x^2 - 2 - \frac{4}{x^3}$

4 Rewrite x^{-2} as $x^{\boxed{?}}$ before differentiating.

5 $\frac{d}{dx}\left(5x^2 - 8 + \frac{4}{x}\right) = \frac{d}{dx}(5x^2 - 8 + 4x^{-1})$
 $= 10x - 4x^{-2}$
 $= 10x - \frac{4}{x^2}$

Rewrite x^{-1}

6 $\frac{d}{dx}\left(\frac{1}{x^4}\right) = \frac{d}{dx}(x^{-4})$
 $= -4x^{-5}$
 $= -\frac{4}{x^5}$

7 $\frac{d}{dx}\left(\frac{2}{x^3}\right) = \frac{d}{dx}(2x^{-3})$
 $= -6x^{-4}$
 $= -\frac{6}{x^4}$

8 $\frac{d}{dx}(3x^{-2}) = -6x^{-3} = -\frac{6}{x^3}$

9 $\frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}\left(x^{\frac{1}{2}}\right)$
 $= \frac{1}{2}x^{-\frac{1}{2}}$
 $= \frac{1}{2x^{\frac{1}{2}}}$
 $= \frac{1}{2\sqrt{x}}$

Rewrite $\sqrt{x}$ as $x^{\boxed{?}}$

10 Let $y = \frac{x}{2-x} = \frac{u}{v}$
 $\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$
 $= \frac{(2-x) \cdot 1 - x(-1)}{(2-x)^2}$
 $= \frac{2-x+x}{(2-x)^2}$
 $= \frac{2}{(2-x)^2}$

10 Division is occurring. What differentiation rule should be used?

$$11 \quad \frac{d}{dx}(x-1)^{15} = 15(x-1)^{14} \frac{d}{dx}(x-1) \\ = 15(x-1)^{14}$$

$$12 \quad \frac{d}{dx}(x^2+1)^8 = 8(x^2+1)^7 \frac{d}{dx}(x^2+1) \\ = 8(x^2+1)^7 \cdot 2x \\ = 16x(x^2+1)^7$$

$$13 \quad \frac{d}{dx}[3(2-x)^5] = 3(5)(2-x)^4 \cdot \frac{d}{dx}(2-x) \\ = 15(2-x)^4(-1) \\ = -15(2-x)^4$$

$$14 \quad y = x^2 + \frac{2}{x} - 1 \\ = x^2 + 2x^{-1} - 1 \\ y' = 2x - 2x^{-2} \\ = 2x - \frac{2}{x^2}$$

$$\text{At } (1, 2), y' = 2(1) - \frac{2}{1^2} = 2 - 2 = 0$$

Since the gradient is 0, the tangent to the curve at (1, 2) must be parallel to the x-axis through the point with y-value of 2.

$\therefore$ equation of tangent is $y = 2$

$$15 \quad y = x^2$$

$$(i) \quad y' = 2x \\ = 2a \quad \text{at } x = a$$

$$(ii) \quad \text{At } x = a, \quad y = a^2 \\ \therefore \text{ point of contact of tangent to the curve is } (a, a^2)$$

$$\text{Equation of tangent: } \frac{y - y_1}{x - x_1} = m$$

$$\therefore \frac{y - a^2}{x - a} = 2a$$

$$\therefore y - a^2 = 2ax - 2a^2$$

$$\therefore y = 2ax - a^2$$

$$(iii) \quad \text{The tangent intersects the x-axis at } (2, 0)$$

$$\therefore 0 = 2a(2) - a^2$$

$$0 = 4a - a^2$$

$$= a(4 - a)$$

$$\therefore a = 0, 4$$

Rejecting $a = 0$, since the tangent at $x = 0$ is the x-axis, then $a = 4$.

$$16 \quad y = x^3 - 3x + 2 \\ y' = 3x^2 - 3$$

$$\text{At } x = -2, \quad y' = 3(-2)^2 - 3 = 12 - 3 = 9$$

$$\text{The gradient } y' = m \quad \therefore m = 9$$

$$\therefore y = 9x + b$$

$$\text{At } x = -2, \quad y = 0 \quad \therefore 0 = 9(-2) + b$$

$$0 = -18 + b \quad \therefore b = 18$$

$$\text{Hence } m = 9, b = 18$$

11 The function is raised to a power. What differentiation rule is to be used?

Alternative method:

$$\text{e.g. } \frac{y - y_1}{x - x_1} = m$$

$$\frac{y - 2}{x - 1} = 0$$

$$\therefore y - 2 = 0$$

$$\text{i.e. } y = 2$$

The gradient $m = y'$
Point-gradient form

16 Alternative method
Equation of tangent:

$$\frac{y - y_1}{x - x_1} = m$$

$$\frac{y - 0}{x - (-2)} = 9$$

$$y = 9(x + 2)$$

$$y = 9x + 18$$

$$\therefore m = 9, b = 18$$

17 $y = x^2 - x$
 $y' = 2x - 1$

At $x = 2$, $y = 2^2 - 2 = 4 - 2 = 2$
 $y' = 2(2) - 1 = 4 - 1 = 3$

∴ tangent is at (2, 2) with gradient of 3

gradient of normal = $-\frac{1}{3}$, using $m_1 m_2 = -1$

∴ equation of normal: $\frac{y - y_1}{x - x_1} = m$
 $\frac{y - 2}{x - 2} = -\frac{1}{3}$
 $y - 2 = -\frac{1}{3}x - \frac{1}{3}(-2)$
 $= -\frac{1}{3}x + \frac{2}{3}$
 $\therefore y = -\frac{1}{3}x + 2\frac{2}{3}$

normal $\Rightarrow$ perpendicular

18 $y = 3x^2 - \frac{p}{x^2} = 3x^2 - px^{-2}$
 $\therefore y' = 6x + 2px^{-3} = 6x + \frac{2p}{x^3}$

At $x = -1$, $y' = 6(-1) + \frac{2p}{(-1)^3} = -6 - 2p$

Since the curve has a turning point, $y' = 0$

∴ $-6 - 2p = 0$
 $2p = -6$
 $p = -3$

Turning points $\Rightarrow$ stationary points
 What is the value of y' for stationary points?

19 (a) $h(x) = (x - 1)(x^2 - 5)$

At $h(x) = 0$, $(x - 1)(x^2 - 5) = 0$

∴ $x = 1$ or $x^2 = 5$

i.e. $x = 1, \pm\sqrt{5}$

i.e. $x = 1, \sqrt{5}, -\sqrt{5}$

(b) $h(x) = (x - 1)(x^2 - 5) = u \cdot v$

$h'(x) = vu' + uv'$

$= (x^2 - 5) \cdot 1 + (x - 1) \cdot 2x$

$= x^2 - 5 + 2x^2 - 2x$

$= 3x^2 - 2x - 5$

$= (3x - 5)(x + 1)$

$= 0$ for stationary points

∴ $3x - 5 = 0$ and $x + 1 = 0$

i.e. $x = \frac{5}{3}, -1$

At $x = \frac{5}{3}$, $h(x) = \left(\frac{5}{3} - 1\right)\left[\left(\frac{5}{3}\right)^2 - 5\right] = -\frac{40}{27}$

At $x = -1$, $h(x) = (-1 - 1)[(-1)^2 - 5] = 8$

∴ stationary points at $\left(\frac{5}{3}, -\frac{40}{27}\right)$ and $(-1, 8)$

Use the 2nd derivative test for max / min.

Since $h'(x) = 3x^2 - 2x - 5$

∴ $h''(x) = 6x - 2$

$$\text{At } x = \frac{5}{3}, \quad h''(x) = 6\left(\frac{5}{3}\right) - 2$$

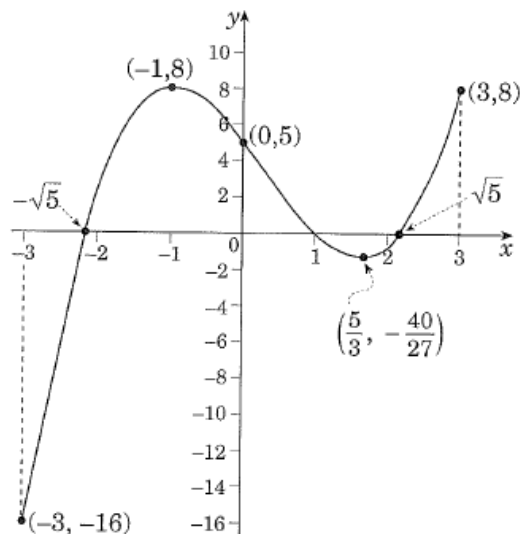
$$= 8 > 0 \quad \therefore \text{a minimum}$$

$$\text{At } x = -1, \quad h''(x) = 6(-1) - 2$$

$$= -8 < 0 \quad \therefore \text{a maximum}$$

i.e. a minimum occurs at $\left(\frac{5}{3}, -\frac{40}{27}\right)$ and a maximum occurs at $(-1, 8)$

(c)



(c) Don't forget to find the y-values at $x = -3$ and 3 .

At the extremities of the domain,

$$\text{for } x = -3, \quad h(x) = (-3-1)[(-3)^2-5]$$

$$= (-4)(4)$$

$$= -16$$

$$\text{for } x = 3, \quad h(x) = (3-1)(3^2-5)$$

$$= (2)(4)$$

$$= 8$$

for the y-axis, $x = 0$,

$$\therefore h(x) = (0-1)(0^2-5)$$

$$= (-1)(-5)$$

$$= 5$$

ALWAYS find the y-values at the domain extremities.

20 $g(x) = x(x+2)^2$

(a) The curve intersects the x-axis at $g(x) = 0$

$$\therefore x(x+2)^2 = 0$$

i.e. $x = 0, -2$

i.e. at $(0, 0), (-2, 0)$ Note: A double root occurs at $(-2, 0)$

(b) $g(x) = x(x+2)^2 = u \cdot v$

$$g'(x) = vu' + uv'$$

$$= (x+2)^2(1) + x \cdot 2(x+2)$$

$$= (x+2)[(x+2) + 2x]$$

$$= (x+2)(3x+2)$$

$$= 0 \text{ for stationary points}$$

$$\therefore x+2 = 0 \text{ and } 3x+2 = 0$$

$$\text{i.e. } x = -2, -\frac{2}{3}$$

$$\text{At } x = -2, \quad g(x) = (-2)(-2+2)^2 = 0$$

Use the product rule or expand out fully, then differentiate.

At $x = -\frac{2}{3}$, $g(x) = \left(-\frac{2}{3}\right)\left(-\frac{2}{3} + 2\right)^2 = -\frac{32}{27}$

$\therefore$ Stationary points occur at $(-2, 0)$ and $\left(-\frac{2}{3}, -\frac{32}{27}\right)$

Use the 2nd derivative test for max / min.

$$g'(x) = (x+2)(3x+2) \\ = 3x^2 + 8x + 4$$

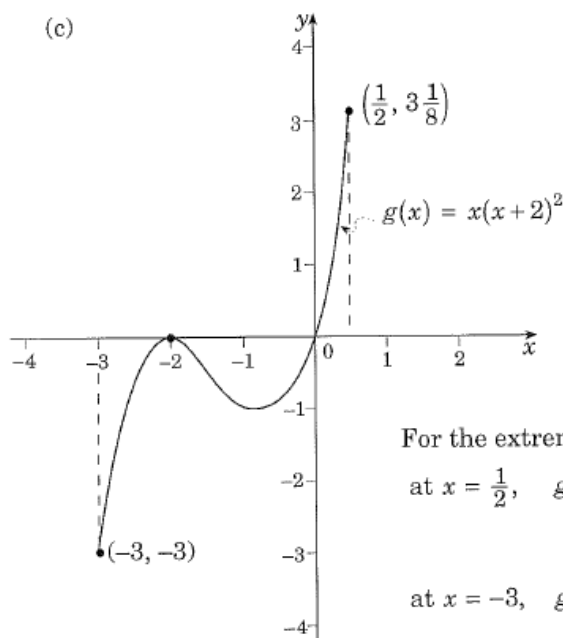
$\therefore g''(x) = 6x + 8$

At $x = -2$, $g''(x) = 6(-2) + 8 \\ = -4 < 0 \therefore$ a maximum

At $x = -\frac{2}{3}$, $g''(x) = 6\left(-\frac{2}{3}\right) + 8 \\ = 4 > 0 \therefore$ a minimum

$\therefore$ a maximum occurs at $(-2, 0)$ and a minimum occurs at $\left(-\frac{2}{3}, -\frac{32}{27}\right)$

(c)



For the extremities of the domain:

at $x = \frac{1}{2}$, $g(x) = \left(\frac{1}{2}\right)\left(\frac{1}{2} + 2\right)^2 \\ = 3\frac{1}{8}$

at $x = -3$, $g(x) = (-3)(-3 + 2)^2 \\ = -3$

21 $\frac{dy}{dx} = x^2(x-2)(x+1) \\ = x^2(x^2 - x - 2) \\ = x^4 - x^3 - 2x^2$

Since $\frac{d^2y}{dx^2} = 4x^3 - 3x^2 - 4x$ involves all terms,

then at $x = 0$, $\frac{d^2y}{dx^2} = 0$, and the 2nd derivative will not indicate the nature of the stationary point.

Consider values close to $x = 0$.

x	< 0	0	> 0
$\frac{dy}{dx}$	$-$	0	$-$

$\Rightarrow$

$\therefore$ a point of horizontal inflexion occurs at $x = 0$

22 (i) $y = 1 - 12x + x^3$
 $y' = -12 + 3x^2$
 $= 3(x^2 - 4)$
 $= 3(x+2)(x-2)$
 $= 0$ for stationary points

$\therefore x = -2, 2$
 $y = 17, -15$

i.e. stationary points occur at $(-2, 17)$ and $(2, -15)$

$y'' = 6x$

At $x = -2$, $y'' = 6(-2)$
 $= -12 < 0 \therefore$ a maximum

At $x = 2$, $y'' = 6(2)$
 $= 12 > 0 \therefore$ a minimum

i.e. a minimum occurs at $(2, -15)$ and a maximum occurs at $(-2, 17)$

(ii) $y' = -12 + 3x^2$
 $y'' = 6x = 0$ for points of inflexion

$\therefore x = 0$

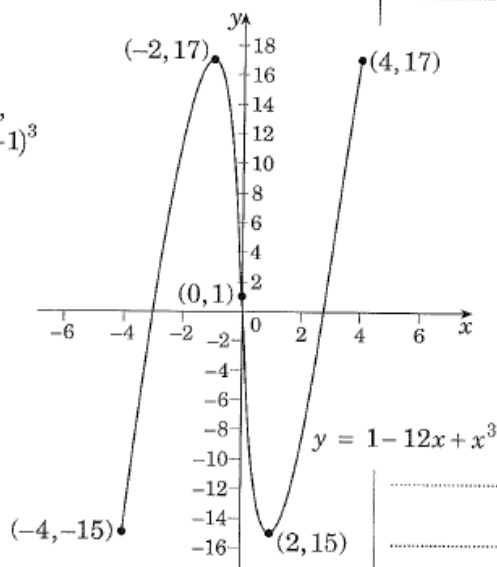
At $x = 0$, $y = 1$

$\therefore$ point of inflexion at $(0, 1)$

(iii) At the extremities of the domain,
for $x = -1$, $y = 1 - 12(-1) + (-1)^3$
 $= 1 + 12 - 1$
 $= 12$

for $x = 4$, $y = 1 - 12(4) + 4^3$
 $= 1 - 48 + 64$
 $= 17$

(iv) The minimum value of the curve is $y = -15$ and it occurs at $x = 2$ (a local minimum turning point) and at $x = -4$ (the extremity of the domain).



23 (i) $y = 6x^2 - 2x^3$
 $\frac{dy}{dx} = 12x - 6x^2$
 $= 6x(2 - x)$
 $= 0$ for stationary points

$\therefore x = 0, 2$
 $y = 0, 8$

i.e. at $(0, 0)$ and $(2, 8)$

$\frac{d^2y}{dx^2} = 12 - 12x$

at $x = 0$, $\frac{d^2y}{dx^2} = 12 - 12(0)$
 $= 12 > 0 \therefore$ a minimum

at $x = 2$, $\frac{d^2y}{dx^2} = 12 - 12(2)$
 $= -24 < 0 \therefore$ a maximum

i.e. a minimum occurs at $(0, 0)$ and a maximum occurs at $(2, 8)$

(ii) The curve intersects the x -axis when $y = 0$

$$\therefore 6x^2 - 2x^3 = 0$$

$$2x^2(3-x) = 0$$

$$\therefore x = 0, 3$$

Also, $\frac{d^2y}{dx^2} = 12 - 12x = 0$ for points of inflexion

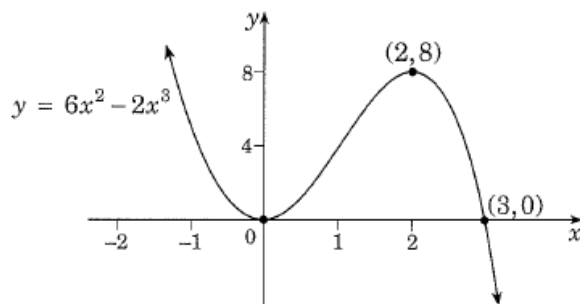
$$\therefore 12 - 12x = 0$$

$$12(1-x) = 0$$

$$\text{i.e. } x = 1$$

$$\text{At } x = 1, y = 6(1^2) - 2(1)^3 = 4$$

i.e. point of inflexion occurs at $(1, 4)$



(iii) Gradient of tangent $\left(\frac{dy}{dx}\right)$ occurs at $(1, 4)$

$$\frac{dy}{dx} = 12x - 6x^2$$

$$= 12(1) - 6(1^2)$$

$$= 6$$

$$\text{Equation of tangent: } \frac{y - y_1}{x - x_1} = m$$

$$\frac{y - 4}{x - 1} = 6$$

$$\therefore y - 4 = 6(x - 1)$$

$$y = 6x - 6 + 4$$

$$\text{i.e. } y = 6x - 2$$

$$m = \frac{dy}{dx}$$

24 (i) $y = \frac{1}{3}x^3 - x^2$

$$\frac{dy}{dx} = \frac{1}{3} \cdot 3x^2 - 2x$$

$$= x^2 - 2x$$

$$= x(x - 2)$$

$$= 0 \text{ for stationary points}$$

$$\therefore x = 0, 2$$

$$\text{At } x = 0, y = \frac{1}{3}(0^3) - 0^2 = 0$$

$$\text{At } x = 2, y = \frac{1}{3}(2^3) - 2^2 = 2\frac{2}{3} - 4 = -1\frac{1}{3}$$

$\therefore$ Stationary points at $(0, 0)$, $(2, -1\frac{1}{3})$

$$\frac{d^2y}{dx^2} = 2x - 2$$

$$\text{at } x = 0, \frac{d^2y}{dx^2} = 2(0) - 2 = -2 < 0 \quad \therefore \text{a maximum}$$

$$\text{at } x = 2, \frac{d^2y}{dx^2} = 2(2) - 2 = 2 > 0 \quad \therefore \text{a minimum}$$

i.e. a maximum occurs at $(0, 0)$ and a minimum occurs at $(2, -1\frac{1}{3})$

(ii) $\frac{d^2y}{dx^2} = 2x - 2 = 0$ for points of inflexion

$$\therefore 2x - 2 = 0$$

$$2x = 2$$

$$x = 1$$

$$\text{At } x = 1, \quad y = -\frac{1}{3}(1^2) - 1^2 = -\frac{2}{3}$$

$$\therefore \text{ a point of inflexion occurs at } \left(1, -\frac{2}{3}\right)$$

(iii) At the extremities of the domain

$$\text{at } x = -\frac{1}{2}, \quad y = \frac{1}{3}\left(-\frac{1}{2}\right)^3 - \left(-\frac{1}{2}\right)^2$$

$$= -\frac{1}{24} - \frac{1}{4}$$

$$= -\frac{7}{24}$$

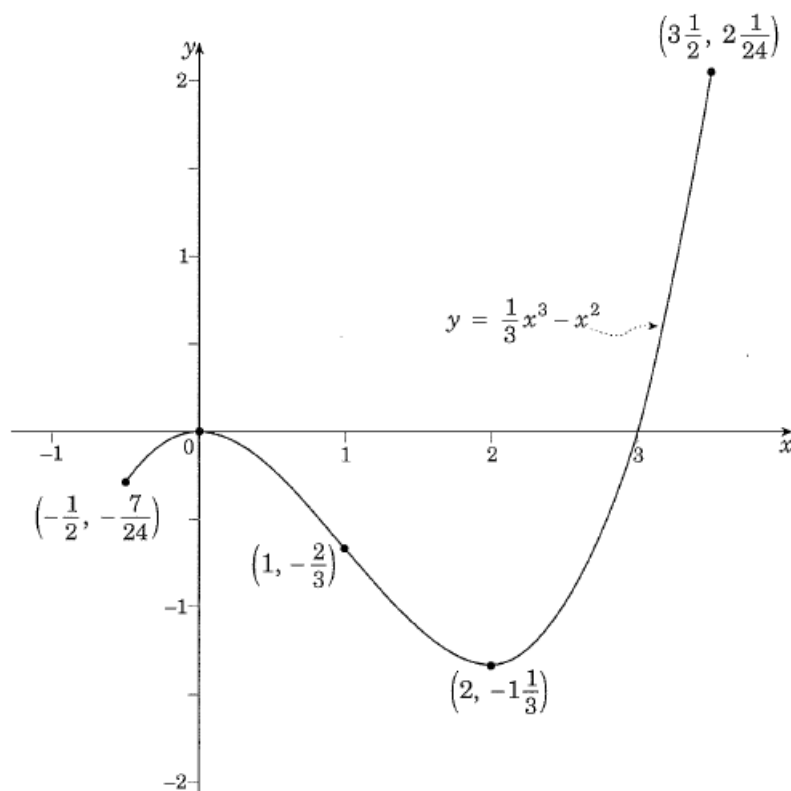
$$\text{at } x = 3\frac{1}{2}, \quad y = \frac{1}{3}\left(3\frac{1}{2}\right)^3 - \left(3\frac{1}{2}\right)^2$$

$$= \frac{1}{3}\left(\frac{7}{2}\right)^3 - \left(\frac{7}{2}\right)^2$$

$$= \frac{343}{24} - \frac{49}{4}$$

$$= \frac{49}{24}$$

$$= 2\frac{1}{24}$$



(iv) Concavity downwards occurs for $\frac{d^2y}{dx^2} < 0$

$$\text{i.e. } 2x - 2 < 0$$

$$2x < 2$$

$$x < 1$$

Thus the curve is concave down for $x < 1$

25 $P = -t^3 + 6t^2 + 16$

(i) $\frac{dP}{dt} = -3t^2 + 12t$
 $= -3t(t-4) = 0$ for stationary values

$\therefore t = 0, 4$

When $t = 0$, $P = 0^3 + 6(0^2) + 16 = 16$

When $t = 4$, $P = (-4)^3 + 6(4^2) + 16$
 $= -64 + 96 + 16$
 $= 48$

$\frac{d^2P}{dt^2} = -6t + 12$

When $t = 0$, $\frac{d^2P}{dt^2} = -6(0) + 12$
 $= 12 > 0 \therefore$ a minimum

When $t = 4$, $\frac{d^2P}{dt^2} = -6(4) + 12$
 $= -12 < 0 \therefore$ a maximum

$\therefore$ maximum population occurs at $t = 4$ years

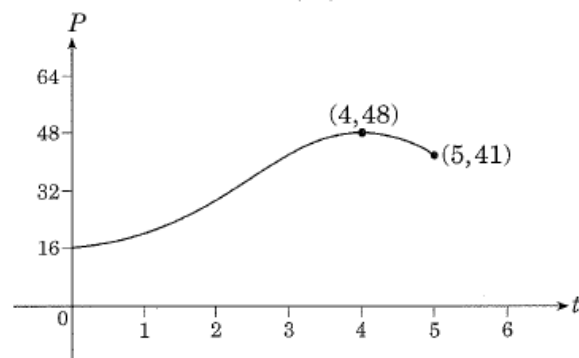
(ii) At $t = 4$, $P = (-4)^3 + 6(4^2) + 16$
 $= -64 + 96 + 16$
 $= 48$

$\therefore$ maximum population was 48 koalas

(iii) For the extremities of the domain,

At $t = 0$, $P = 0^3 + 6(0)^2 + 16 = 16$

At $t = 5$, $P = (-5)^3 + 6(5^2) + 16 = 41$



(iv) The curve $P = -t^3 + 6t^2 + 16$ has a point of inflexion when $\frac{d^2P}{dt^2} = 0$

$\therefore -6t + 12 = 0$

$\therefore 6t = 12$

$t = 2$

Since the curve is concave down for $t > 2$ in the domain

$0 \leq t \leq 5$, the most rapid increase must occur at $t = 2$ years.

26 (a) Surface area = area_{base} + 2 × area_{side} + 2 × area_{front}

$\therefore A = 6x^2 + 2(2xy) + 2(3xy)$

i.e. $A = 6x^2 + 10xy$

Volume: $V = (3x)(2x)y = 6x^2y$

(b) $240 = 6x^2 + 10xy$

$\therefore 10xy = 240 - 6x^2$

$y = \frac{240 - 6x^2}{10x}$

continued ...

Be careful of signs

Substituting into $V = 6x^2y$, we have

$$\begin{aligned} V(x) &= 6x^2 \left(\frac{240 - 6x^2}{10x} \right) \\ &= \frac{3x(240 - 6x^2)}{5} \\ &= \frac{18x(40 - x^2)}{5} \end{aligned}$$

(c) Since $V(x) > 0$, then $\frac{18x(40 - x^2)}{5} > 0$

Since $x > 0$, then $40 - x^2 > 0$

$$40 > x^2$$

$$x^2 < 40$$

$$x < \sqrt{40} = 2\sqrt{10}$$

i.e. $x < 2\sqrt{10}$

(d) $V = \frac{18x(40 - x^2)}{5} = \frac{18}{5}(40x - x^3)$

$$\frac{dV}{dx} = \frac{18}{5}(40 - 3x^2) = 0 \text{ for stationary values}$$

$$\therefore 40 - 3x^2 = 0$$

$$3x^2 = 40$$

$$x^2 = \frac{40}{3}$$

$$\therefore x = \sqrt{\frac{40}{3}} \text{ since } x > 0$$

$$\frac{d^2V}{dx^2} = \frac{18}{5}(-6x)$$

$$= -\frac{18}{5} \cdot 6 \sqrt{\frac{40}{3}}$$

$$< 0 \quad \therefore V \text{ is a maximum at } x = \sqrt{\frac{40}{3}}$$

$$\text{When } x = \sqrt{\frac{40}{3}}, \quad V(x) = 18\sqrt{\frac{40}{3}} \left[\frac{40 - \left(\sqrt{\frac{40}{3}}\right)^2}{5} \right]$$

$$= 18 \times \frac{2\sqrt{10}}{\sqrt{3}} \left(\frac{40 - \frac{40}{3}}{5} \right)$$

$$= \frac{36\sqrt{10}\sqrt{3}}{3} \left(\frac{120 - 40}{5} \right)$$

$$= 12\sqrt{30} \times \frac{80}{15}$$

$$= 64\sqrt{30}$$

27 $A = 5t^2 - t^3$

(i) $\frac{dA}{dt} = 10t - 3t^2 = t(10 - 3t) = 0$ for stationary values

$$\therefore t = 0 \text{ and } 10 - 3t = 0$$

i.e. $3t = 10 \quad \therefore t = \frac{10}{3} \text{ hours} = 3\frac{1}{3} \text{ hours}$

Rationalise the denominator for $18 \times \frac{2\sqrt{10}}{\sqrt{3}} \rightarrow 18 \times \frac{2\sqrt{10}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{36\sqrt{10}\sqrt{3}}{3}$

When $t = 0$, $A = 0$

$$\begin{aligned}\text{When } t = 3\frac{1}{3}, \quad A &= 5\left(\frac{10}{3}\right)^2 - \left(\frac{10}{3}\right)^3 \\ &= \frac{500}{9} - \frac{1000}{27} \\ &= \frac{1500}{27} - \frac{1000}{27} \\ &= \frac{500}{27} \\ &= 18\frac{14}{27}\end{aligned}$$

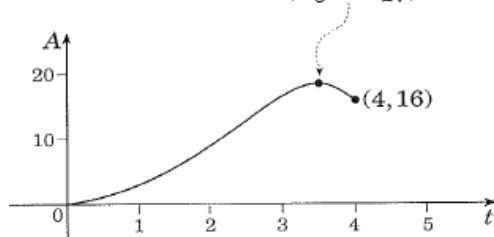
At the extremity, $x = 4$,

$$\begin{aligned}\therefore A &= 5(4)^2 - (4)^3 \\ &= 80 - 64 \\ &= 16\end{aligned}$$

$$\text{Also, } \frac{d^2A}{dt^2} = 10 - 6t$$

$$\begin{aligned}\text{At } t = \frac{10}{3}, \quad \frac{d^2A}{dt^2} &= 10 - 6\left(\frac{10}{3}\right) \\ &= -20 < 0 \quad \therefore \text{a maximum}\end{aligned}$$

i.e. a maximum occurs at $\left(3\frac{1}{3}, 18\frac{14}{27}\right)$



(ii) The maximum amount occurs at $t = 3\frac{1}{3}$ hours.

(iii) The most rapid increase occurs when the curve changes concavity

i.e. at the point of inflexion when $\frac{d^2A}{dt^2} = 0$

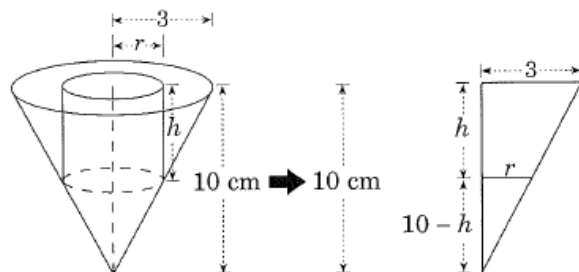
$$\frac{d^2A}{dt^2} = 10 - 6t = 0$$

$$\therefore 6t = 10$$

$$t = \frac{10}{6} = \frac{5}{3} = 1\frac{2}{3} \text{ hours}$$

$\therefore$ the most rapid increase occurs at $t = 1\frac{2}{3}$ hours

28 (i)



Redrawing the diagram as a half side view, we obtain a pair of similar triangles.

$$\therefore \frac{10-h}{r} = \frac{10}{3}$$

$$10-h = \frac{10r}{3}$$

$$\therefore h = 10 - \frac{10r}{3}$$

$$\text{Volume (cylinder)} = \pi r^2 h$$

$$\text{i.e. } V = \pi r^2 \left(10 - \frac{10r}{3} \right)$$

$$= \pi r^2 \left(\frac{30 - 10r}{3} \right)$$

$$\therefore V = \frac{10\pi r^2(3-r)}{3}$$

$$(ii) \quad V = \frac{10\pi(3r^2 - r^3)}{3}$$

$$\frac{dV}{dr} = \frac{10\pi}{3}(6r - 3r^2)$$

$$= 10\pi(2r - r^2)$$

$$= 10\pi r(2 - r)$$

$$= 0 \text{ for stationary values}$$

$$\therefore r = 0, 2$$

$$\text{Rejecting } r = 0 \text{ since } r > 0, \quad \therefore r = 2 \text{ cm}$$

$$\text{Using the 2nd derivative test: } \frac{d^2V}{dr^2} = 10\pi(2 - 2r)$$

$$\text{When } r = 2, \quad \frac{d^2V}{dr^2} = 10\pi(-2) = -20\pi < 0$$

$$\therefore V \text{ is a maximum when } r = 2 \text{ cm}$$

$$\text{Also, } h = 10 - \frac{10(2)}{3}$$

$$= \frac{30 - 20}{3}$$

$$= \frac{10}{3} \text{ cm}$$

$$= 3\frac{1}{3} \text{ cm}$$

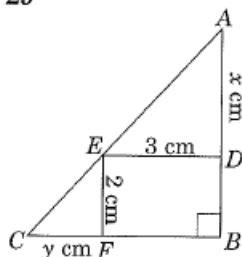
$$\therefore \text{Maximum volume occurs at } r = 2 \text{ cm, } h = 3\frac{1}{3} \text{ cm.}$$

$$(iii) \quad V_{\max} = \pi r^2 h$$

$$= \pi(2^2) \frac{10}{3}$$

$$= \frac{40\pi}{3} \text{ cm}^3$$

29



- (i) Since $BDEF$ is a rectangle,
 $ED \parallel FB$, $EF \parallel BD \parallel BA$, and
 $\angle FBD = \angle EDA = 90^\circ = \angle EFC$

For the Δ s EFC , ADE ,

$$\angle ADE = \angle EFC \quad (\text{both } 90^\circ)$$

$$\angle DAE = \angle FEC \quad (\text{corresp. } \angle\text{s with } EF \parallel AD)$$

$$\angle DEA = \angle FCE \quad (\text{corresp. } \angle\text{s with } ED \parallel CF)$$

$$\therefore \Delta EFC \parallel \Delta ADE \quad (\text{corresp. equal } \angle\text{s})$$

- (ii) Since Δ s EFC , ADE are similar,

$$\frac{AD}{EF} = \frac{ED}{CF} \quad (\text{corresp. sides are in the same ratio})$$

$$\therefore \frac{x}{2} = \frac{3}{y}$$

$$\text{i.e. } xy = 6$$

$$(iii) \text{ Area } (\triangle ABC) = \frac{1}{2} \cdot BC \cdot AB$$

$$\begin{aligned} A &= \frac{1}{2}(y+3)(x+2) \\ &= \frac{1}{2}(xy+2y+3x+6) \\ &= \frac{1}{2}\left(6+2 \cdot \frac{6}{x}+3x+6\right) \\ &= \frac{1}{2}\left(12+\frac{12}{x}+3x\right) \\ &= \frac{3x}{2}+\frac{6}{x}+6 \end{aligned}$$

$$(iv) \quad A = \frac{3x}{2} + 6x^{-1} + 6$$

$$\begin{aligned} \frac{dA}{dx} &= \frac{3}{2} - 6x^{-2} \\ &= \frac{3}{2} - \frac{6}{x^2} = 0 \text{ for stationary values} \end{aligned}$$

$$\therefore \frac{3}{2} = \frac{6}{x^2}$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

Rejecting $x = -2$ since $x > 0$, then $x = 2$.

$$\frac{d^2A}{dx^2} = 6x^{-3} = \frac{6}{x^3}$$

$$\text{When } x = 2, \quad \frac{d^2A}{dx^2} = \frac{6}{2^3} = \frac{6}{8} > 0$$

$\therefore A$ is a minimum for $x = 2$ cm

Height $AB = x + 2 = 2 + 2 = 4$ cm

Base $BC = y + 3 = 3 + 3 = 6$ cm

$$30 \text{ (i) Time of flight} = \frac{\text{distance}}{\text{speed}} = \frac{3000}{v} \text{ hours.}$$

Amount of fuel (A) = rate $\times$ time

$$\begin{aligned} \text{i.e. } A &= (50 + 10^{-6}v^3) \times \frac{3000}{v} \\ &= \frac{150\,000}{v} + \frac{3000 \times 10^{-6}v^3}{v} \\ &= \frac{150\,000}{v} + \frac{3v^2}{1000} \end{aligned}$$

$$(ii) \quad A = 150\,000v^{-1} + \frac{3v^2}{1000}$$

$$\frac{dA}{dv} = -150\,000v^{-2} + \frac{3(2v)}{1000} = 0 \text{ for stationary values}$$

$$\therefore \quad 150\,000v^{-2} = \frac{6v}{1000}$$

$$\frac{6v}{1000} = \frac{150\,000}{v^2}$$

$$v^3 = \frac{150\,000 \times 1000}{6} = 25\,000\,000$$

$$\therefore \quad v = \sqrt[3]{25\,000\,000}$$

$$= 292.40 \text{ km/h}$$

$$= 292 \text{ km/h (3 sig. figs)}$$

$$\begin{aligned} BC &= CF + FB = y + 3 \\ AB &= AD + DB = x + 2 \end{aligned}$$

$$xy = 6 \Rightarrow y = \frac{6}{x}$$

$$y = \frac{6}{x} = \frac{6}{2} = 3$$

$$\frac{d^2 A}{dv^2} = 2 \times 150\,000v^{-3} + \frac{6}{1000} > 0 \text{ for all } v \text{ since } v > 0$$

$\therefore A$ is a minimum when $v = 292$

$$\begin{aligned} A &= \frac{150\,000}{292} + \frac{3(292)^2}{1000} \\ &= 769.49 \text{ litres} \\ &= 769 \text{ L (3 sig. figs)} \end{aligned}$$

Minimum fuel used is equivalent to greatest fuel economy.

31 (i) $AC = 5 - x$

(ii) By Pythagoras' theorem,

$$BC^2 + AB^2 = AC^2$$

$$BC^2 + x^2 = (5 - x)^2$$

$$\begin{aligned} \therefore BC^2 &= (5 - x)^2 - x^2 \\ &= 25 - 10x + x^2 - x^2 \\ &= 25 - 10x \end{aligned}$$

$$BC = \sqrt{25 - 10x}$$

$$\text{Area}(\triangle ABC) = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times (\sqrt{25 - 10x}) \times x$$

$$\text{i.e. Area} = \frac{1}{2} x \sqrt{25 - 10x}$$

$$(iii) A = \frac{1}{2} x (25 - 10x)^{\frac{1}{2}}$$

$$\frac{dA}{dx} = \frac{1}{2} x (-10)(25 - 10x)^{-\frac{1}{2}} + \frac{1}{2} (25 - 10x)^{\frac{1}{2}} (1)$$

$$= \frac{-10x}{2\sqrt{25 - 10x}} + \frac{\sqrt{25 - 10x}}{2}$$

$$= \frac{-10x + 25 - 10x}{2\sqrt{25 - 10x}}$$

$$= \frac{25 - 20x}{2\sqrt{25 - 10x}} = 0 \text{ for stationary values}$$

$$\therefore 25 - 20x = 0$$

$$20x = 25$$

$$x = \frac{25}{20} = 1.25$$

Applying the 2nd derivative test involves a lot of algebraic processes

$\therefore$ test the gradient close to values of $x = 1.25$

x	< 1.25	1.25	> 1.25
$\frac{dA}{dx}$	+	0	-

Slope $\nearrow \frac{0}{\searrow}$

$\therefore A$ is a maximum at $x = 1.25$

$$\text{From (ii), } A = \frac{1}{2} \times \frac{5}{4} \sqrt{25 - 10\left(\frac{5}{4}\right)}$$

$$= \frac{5}{8} \sqrt{25 - \frac{25}{2}}$$

$$= \frac{5}{8} \sqrt{\frac{25}{2}}$$

$$= \frac{5}{8} \times \frac{5}{\sqrt{2}}$$

$$= \frac{25}{8} \times \frac{\sqrt{2}}{2}$$

$$= \frac{25\sqrt{2}}{16} \text{ m}^2$$

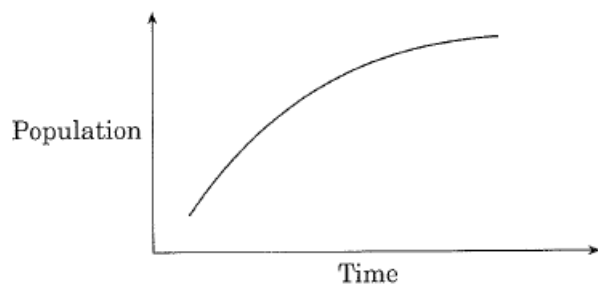
Product rule and chain rule

Bring both terms to a common denominator:

$$\begin{aligned} &\frac{\sqrt{25 - 10x}}{2} \\ &= \frac{\sqrt{25 - 10x} \sqrt{25 - 10x}}{2\sqrt{25 - 10x}} \\ &= \frac{25 - 10x}{2\sqrt{25 - 10x}} \end{aligned}$$

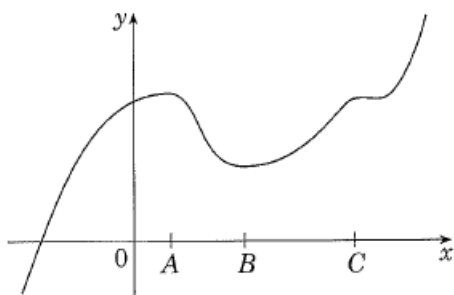
Rationalise the denominator

32

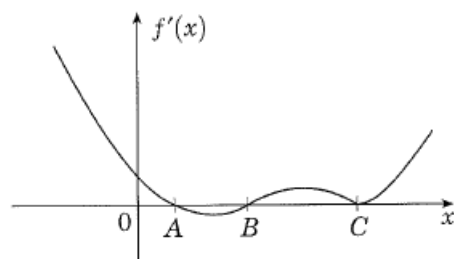


- 33 During the year the interest rate continues to fall, but the rate of change alters more slowly in the second half of the year, tending to become constant.

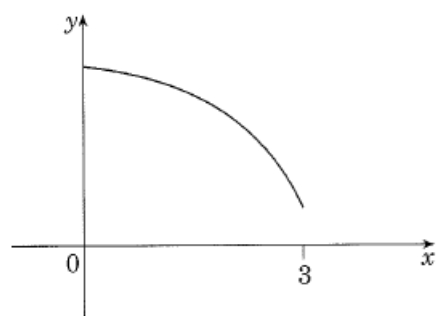
34 (i)



(ii)



35



$f'(x) < 0 \Rightarrow$	
$f''(x) < 0 \Rightarrow$	